

Packet Tracer - Identify Packet Flow

Objectives

In this activity, you will observe packet flow in a LAN and WAN topology. You will also observe how the packet flow path may change when there is a change in the network topology.

Part 1: Verify Connectivity

Part 2: Remote LAN Network Topology

Part 3: WAN Network Topology

Background / Scenario

Packet Tracer allows the design and creation of a simulated networking topology. In this activity, you are presented with a simplified topology to observe packet flow. You will explore how packets travel through the network using the simulation mode in Packet Tracer. You will also observe the changes in packet flow when there is a change in the network topology.

Required Resources

- Latest version of Packet Tracer installed

Instructions

Part 1: Verifying Connectivity

In this part, you will verify that you can access the other networks from devices on the Home Network.

- Click **PC0**. Select the **Desktop** tab and open the **Web Browser**.
- In the URL field, enter **www.cisco.pka** and press **Go**. Be sure to use the .pka domain, not the .com domain. It should be successful. You can click **Fast Forward Time** to speed up the process.
- Repeat this for **www.web.pka**. It should be successful.
- Exit the web browser when finished.

Part 2: Remote LAN Network Topology

In this part, you will use the simulation mode in Packet Tracer to observe how packets flow through a remote LAN network.

- Switch to Simulation mode (Shift + S). Click **Show All/None** to clear all the selected event list filters.
- Click **Edit Filters**. Select **DNS** under the IPv4 tab and **HTTP** under **Misc** tab.
- Open a web browser on **PC0**. Enter **www.web.pka** and press **Go**.
Question: Predict the packet path to resolve **www.web.pka** to an IP address. Record your prediction.

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	DNS
	0.001	PC0	Wireless Router0	DNS
	0.002	Wireless Router0	Cable Modem0	DNS
	0.003	Cable Modem0	Cloud0	DNS
	0.004	Cloud0	Router5	DNS
	0.005	Router5	East	DNS
	0.006	East	Switch0	DNS
	0.007	Switch0	Switch1	DNS
	0.008	Switch1	Public DNS	DNS
	0.009	Public DNS	Switch1	DNS
	0.010	Switch1	Switch0	DNS
	0.011	Switch0	East	DNS
	0.012	East	Router5	DNS
	0.013	Router5	Cloud0	DNS
	0.014	Cloud0	Cable Modem0	DNS
	0.015	Cable Modem0	Wireless Router0	DNS
	0.016	Wireless Router0	PC0	DNS

- d. Click **Capture / Forward** until the webpage is displayed on PC0 to view the packet flow. Click **View Previous Events** when prompted by the Buffer Full dialog box.

After the IP address has been resolved, which path did HTTP packets travel to display the webpage? Record your observations.

0.034	--	PC0	HTTP
0.035	--	PC0	HTTP
0.036	PC0	Wireless Router0	HTTP
0.037	Wireless Router0	Cable Modem0	HTTP
0.038	Cable Modem0	Cloud0	HTTP
0.039	Cloud0	Router5	HTTP
0.040	Router5	East	HTTP
0.041	East	Switch0	HTTP
0.042	Switch0	Switch1	HTTP
0.043	Switch1	Switch2	HTTP
0.044	Switch2	www.web.pka	HTTP
0.045	www.web.pka	Switch2	HTTP
0.046	Switch2	Switch1	HTTP
0.047	Switch1	Switch0	HTTP
0.048	Switch0	East	HTTP
0.049	East	Router5	HTTP
0.050	Router5	Cloud0	HTTP

- e. Switch to Real time mode (Shift + R). Click the X icon in the right tool panel to select the Delete tool. Remove the link between Switch0 and Switch 1 from the Public Network to simulate a broken link. After 30 seconds, the network will learn of the broken link. You can click Fast Forward to speed up the process.
- f. Select the Arrow tool above the Delete tool to de-select Delete.
- g. Switch to Simulation mode (Shift + S). Open a web browser in **Tablet0** and navigate to **www.web.pka**. You can click Auto Capture/Play to have Packet Tracer forward the packets without your interaction. You can also move the Play Slider to the right to speed up the packet forwarding.
- With a broken link in the LAN, how did the path change? Record your observation.

Before Broken Link

PCs communicated through
Switch0 → Switch1 directly

Network topology was a loop
with redundant paths

After Broken Link

PCs now communicate through an alternate
route, Switch0 → Switch2 → Switch1

One path is now down, and STP reconverged
to use the available link

Part 3: WAN Network Topology

Step 1: PC0 to websites.

- Remaining in Simulation mode, open a web browser on **PC0**. Enter **www.cisco.pka** and press **Go**. Predict the packet path to resolve **www.cisco.pka** to an IP address. Record your prediction.

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.341	PC0	Wireless Router0	HTTP
	0.342	Wireless Router0	Cable Modem0	HTTP
	0.343	Cable Modem0	Cloud0	HTTP
	0.344	Cloud0	Router5	HTTP
	0.345	Router5	Router2	HTTP
	0.346	Router2	Router4	HTTP
	0.347	Router4	West	HTTP
	0.348	West	Switch	HTTP
	0.349	Switch	www.cisco.pka	HTTP
	0.350	www.cisco.pka	Switch	HTTP
	0.351	Switch	West	HTTP
	0.352	West	Router4	HTTP
	0.353	Router4	Router2	HTTP
	0.354	Router2	Router5	HTTP
	0.355	Router5	Cloud0	HTTP
	0.356	Cloud0	Cable Modem0	HTTP
	0.357	Cable Modem0	Wireless Router0	HTTP
	0.358	Wireless Router0	PC0	HTTP

- Click **Capture / Forward** until the webpage is displayed on PC0 to view the packet flow. Click **View Previous Events** when prompted by the Buffer Full dialog box.
After the IP address has been resolved, which path did HTTP packets travel to display the webpage? Record your observations.

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.341	PC0	Wireless Router0	HTTP
	0.342	Wireless Router0	Cable Modem0	HTTP
	0.343	Cable Modem0	Cloud0	HTTP
	0.344	Cloud0	Router5	HTTP
	0.345	Router5	Router2	HTTP
	0.346	Router2	Router4	HTTP
	0.347	Router4	West	HTTP
	0.348	West	Switch	HTTP
	0.349	Switch	www.cisco.pka	HTTP
	0.350	www.cisco.pka	Switch	HTTP
	0.351	Switch	West	HTTP
	0.352	West	Router4	HTTP
	0.353	Router4	Router2	HTTP
	0.354	Router2	Router5	HTTP
	0.355	Router5	Cloud0	HTTP
	0.356	Cloud0	Cable Modem0	HTTP
	0.357	Cable Modem0	Wireless Router0	HTTP
	0.358	Wireless Router0	PC0	HTTP

- c. Switch to Real time mode (Shift + R). Remove the link between Router4 and Router2 from the topology to simulate an inaccessible path. The routers are using Enhanced Interior Gateway Routing Protocol (EIGRP) to dynamically adjust routing tables to account for the deleted link.
- d. Switch to Simulation mode (Shift + S). Open a web browser in **Tablet0** and navigate to **www.cisco.pka**.
 With a broken link in the WAN, how would the path change? Record your observation.

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.047	--	Tablet0	HTTP
	0.048	Tablet0	Wireless Router0	HTTP
	0.049	Wireless Router0	Cable Modem0	HTTP
	0.050	Cable Modem0	Cloud0	HTTP
	0.051	Cloud0	Router5	HTTP
	0.052	Router5	Router2	HTTP
	0.053	Router2	Router3	HTTP
	0.053	--	Wireless Router0	HTTP
	0.054	Wireless Router0	Tablet0	HTTP
	0.054	Router3	Router4	HTTP
	0.055	Router4	West	HTTP
	0.056	West	Switch	HTTP
	0.057	Switch	www.cisco.pka	HTTP
	0.058	www.cisco.pka	Switch	HTTP
	0.059	Switch	West	HTTP
	0.060	West	Router4	HTTP
	0.061	Router4	Router3	HTTP
	0.062	Router3	Router2	HTTP

0.063	Router2	Router5	HTTP
0.064	Router5	Cloud0	HTTP
0.065	Cloud0	Cable Modem0	HTTP
0.066	Cable Modem0	Wireless Router0	HTTP
0.067	Wireless Router0	Tablet0	HTTP

- e. Switch to Real time mode (Shift + R).

Step 2: PC1 to websites.

- Click PC1 > Desktop and open a command prompt.
- Enter **tracert www.web.pka** at the command.

PC> **tracert** www.web.pka

```
C:\>tracert www.web.pka

Tracing route to 209.165.202.132 over a maximum of 30 hops:

  1  38 ms    0 ms    0 ms    192.168.0.1
  2   0 ms    0 ms    1 ms    209.165.200.225
  3   1 ms    2 ms    0 ms    192.0.2.2
  4   3 ms    3 ms    4 ms    192.0.2.18
  5  24 ms    3 ms    2 ms    192.0.2.26
  6   2 ms    2 ms    1 ms    209.165.202.132
  7  10 ms    12 ms   10 ms    209.165.202.132

Trace complete.

C:\>
```

- Match the IP addresses in the **tracert** results to the devices in the topology. Hover over the routers in the topology to view the IP addresses of the interfaces on the routers. If the popup does not stay active long enough, you can access router IP addresses in the following manner: Click the router > CLI > press Enter. Now enter the command **show ip interface brief** to getting a listing of the interfaces and IP addresses.

Trace Number	Device	Interface	IP Address
1	WEST	GigabitEthernet0/1	192.168.0.1
2	router4	Serial 0/1/1	209.165.200.225
3	router 3	Serial0/0/0	192.0.2.2
4	router2	Serial0/0/1	192.0.2.18
5	router5	Serial 0/1/1	192.0.2.26
6	East	Serial 0/0/0	209.165.202.130
7	www.web.pka	NIC	209.165.202.132 / 192.168.2.254

Blank Line. No additional information

Network address translation (NAT) is used to translate the private www.web.pka IP address of 192.168.2.254 to a routable IPv4 address of 209.165.202.132. In the tracert result, the first line of IPv4

address of 209.165.202.132 is for the G0/1 interface of East. The second line of IPv4 address of 209.165.202.132 displays the public IPv4 address of the web server.

- d. Switch to Simulation mode (Shift + S). Open the web browser on PC1 and enter www.web.pka as the URL. Click **Go**.
- e. Click **Capture / Forward** to load the web page.

Compare the **tracert** results to the simulation results for the HTTP packets. Record your observations.

HTTP packets result

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.044	PC1	Switch	HTTP
	0.045	Switch	West	HTTP
	0.046	West	Router4	HTTP
	0.047	Router4	Router3	HTTP
	0.048	Router3	Router2	HTTP
	0.049	Router2	Router5	HTTP
	0.050	Router5	East	HTTP
	0.051	East	Switch0	HTTP
	0.052	Switch0	Switch2	HTTP
	0.053	Switch2	www.web.pka	HTTP
	0.054	www.web.pka	Switch2	HTTP
	0.055	Switch2	Switch0	HTTP
	0.056	Switch0	East	HTTP
	0.057	East	Router5	HTTP
	0.058	Router5	Router2	HTTP
	0.059	Router2	Router3	HTTP
	0.060	Router3	Router4	HTTP
	0.061	Router4	West	HTTP
	0.062	West	Switch	HTTP
	0.063	Switch	PC1	HTTP

Tracert result

```

C:\>tracert www.web.pka

Tracing route to 209.165.202.132 over a maximum of 30 hops:

  1  38 ms    0 ms    0 ms    192.168.0.1
  2   0 ms    0 ms    1 ms    209.165.200.225
  3   1 ms    2 ms    0 ms    192.0.2.2
  4   3 ms    3 ms    4 ms    192.0.2.18
  5  24 ms    3 ms    2 ms    192.0.2.26
  6   2 ms    2 ms    1 ms    209.165.202.132
  7  10 ms    12 ms   10 ms    209.165.202.132

Trace complete.

C:\>

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